Practice workbook · Finding the mode

Finding the most frequent value, or stating the procedure for finding it.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-RM-17

1. Question 4 · part a · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part a only.

4 A school is implementing a ban on unhealthy food.

Davide is interested to see whether young people would change their attitude towards eating unhealthy food due to the ban. 10 participants aged 11 to 16 years old from the school were asked to rate whether you should eat unhealthy food, using a scale of 1 to 10, with 1 being 'should never eat it' and 10 being 'should eat it every day'.

Davide tested the participants two months before the ban and again two months after the ban was put in place.

The results of Davide's study are shown in **Table 4**.

Participant	Attitude towards eating unhealthy food before the ban (1–10)	Attitude towards eating unhealthy food after the ban (1–10)
A	8	6
B	7	6
C	7	4
D	4	1
E	8	5
F	9	5
G	7	5
H	6	4
I	7	3
J	10	7

Table 4

(a) Calculate the mode for attitude towards eating unhealthy food **before** the ban.

(1)

Mode .....

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10

P 7 8 7 3 4 A 0 1 0 5 2

1. Question 4 · part a · 1 mark

2025 JUN · Paper 02

Worked-example retry · Attempt part a only.

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(b) Explain **one** conclusion that Davide could make about attitudes towards eating unhealthy food after the ban was put in place from the results in **Table 4**.

(2)

(c) Davide decides to conduct further research gathering qualitative data.

Define what is meant by 'qualitative data'. Use an example in your answer.

(2)

(Total for Question 4 = 5 marks)

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1. Question 4 · part a · 1 mark · Working and answer

Working and answer

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2. Question 4 · part whole · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part whole only.

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4 State how you would find the mode in a set of data.

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(Total for Question 4 = 1 mark)

Working and answer

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3. Question 3 · part d · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part d only.

3 Juliet investigated whether there was a relationship between the number of hours spent using social media and self-rated body image scores.

Firstly, Juliet interviewed a sample of 17-year-old students using a structured interview method, where she asked them to state the length of time they spent on social media per day.

Juliet then asked the same participants to complete a questionnaire to rate how happy they were with their appearance, with a score of 0 being 'very unhappy' and 10 being 'very happy'.

(a) Define what is meant by a 'structured interview'.

(1)

(b) Describe how Juliet could have used a random sampling technique for her investigation.

(2)

(c) Identify the type of data Juliet gathered in her investigation.

(1)

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P 7 0 4 8 6 A 0 8 4 8

3. Question 3 · part d · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part d only.

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The results of Juliet's investigation are shown in **Table 3**.

Participant	Number of hours per day using social media	Body image score (out of 10)
A	2	6
B	1	7
C	4	3
D	6	2
E	2	7
F	5	3
G	1	9
H	2	7
I	7	1

Table 3

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(d) Calculate the mode for body image scores.

(1)

Mode .....

(e) Juliet concludes that there is a negative relationship between social media use and body image.

Give **one** way Juliet's findings could support this conclusion.

(1)

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(Total for Question 3 = 6 marks)

3. Question 3 · part d · 1 mark · Working and answer

Working and answer

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4. Question 3 · part c · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c only.

3 Quentin is investigating the capacity of short-term memory. He showed 20 objects to 10 participants for 30 seconds. Quentin then asked them to recall as many as they could.

His results are shown in **Table 2**.

Participant	Number of objects recalled
A	10
B	9
C	8
D	8
E	6
F	7
G	8
H	9
I	10
J	11

Table 2

(a) Calculate how many participants recalled nine objects as a fraction of all participants.

You must give your answer in the lowest form.

(1)

Fraction .....

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4. Question 3 · part c · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c only.

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(b) Calculate the median for the number of objects recalled.

(2)

Median .....

(c) Calculate the mode for the number of objects recalled.

(1)

Mode .....

(d) State how Quentin would calculate the mean score for the number of objects recalled.

(1)

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(Total for Question 3 = 5 marks)

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4. Question 3 · part c · 1 mark · Working and answer

Working and answer

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5. Question 5 · part d · 1 mark

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Further practice · Attempt part d only.

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5 Rocio conducted a case study on her patient who had an addiction to alcohol. She worked with the patient for six months to help treat their addiction.

(a) Define what is meant by a 'case study'.

(1)

(b) Rocio recorded the conversations with the patient as part of her case study. She also monitored the alcohol consumption of the patient in units of alcohol (one unit of alcohol = 10 ml).

Her results are shown in **Table 2**.

Month	Total alcohol consumption in units
March	298
April	221
May	181
June	140
July	99
August	59

Table 2

Describe, using an example from Rocio's case study, what is meant by 'quantitative data'.

(2)

5. Question 5 · part d · 1 mark

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Further practice · Attempt part d only.

(c) Estimate the number of units of alcohol that the patient has reduced their consumption by between March and August.

(1)

Estimation .....

(d) For the months of March and April, Rocio recorded the patient's alcohol consumption each week. Her results are shown in **Table 3**.

Week	Total alcohol consumption in units
March Week 1	79
March Week 2	77
March Week 3	72
March Week 4	70
April Week 1	79
April Week 2	63
April Week 3	45
April Week 4	34

Table 3

Calculate the mode for the units of alcohol consumed by the patient using **Table 3**.

(1)

Mode .....

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P 7 5 5 2 4 A 0 1 0 4 8

5. Question 5 · part d · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part d only.

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(e) Calculate the median score for the units of alcohol consumed by the patient using **Table 3**.

(2)

Median .....

(f) On one day in March, the patient drank a total of 12.81924 units of alcohol.

Give the units of alcohol to **two** decimal places.

(1)

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(Total for Question 5 = 8 marks)

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Working and answer